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Enhancing Privacy, Security, and Performance in IoT Ecosystems Using AI-Powered Encoding Mechanisms

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Abstract

The Internet of Things (IoT) ecosystem faces significant privacy and security challenges due to its heterogeneous devices, dynamic behavior, and evolving hacking techniques. Traditional approaches struggle to provide a unified solution. To address these issues, this study proposes an integrated privacy and security mechanism using a novel encoding technique known as Replacement Encoding (RE). Designed to protect sensitive information during AI model training, RE camouflages data while preserving model integrity and utility. Additionally, it offers automated data preprocessing to enhance AI performance.

Message packet features were extracted from the CICIoT2023 dataset (PCAP files) using Wireshark for analysis. These features were processed with the proposed RE technique and integrated into AI classifiers for attack detection. The RE-based models achieved accuracies of 88.94% using Random Forest (RF) and 86.61% using Deep Neural Networks (DNN), employing 100 features. In comparison, models utilizing genetic algorithm-based correlation features with up to 15 features reached accuracies of 90.16% (RF) and 94.81% (DNN), indicating the potential of optimized feature selection.

The study demonstrates the RE mechanism's effectiveness not only in maintaining privacy and enhancing preprocessing but also in securing sensitive data. Its applicability extends to various domains, including Generative Pre-Training Transformer (GPT) systems and other AI applications. Overall, this unified framework contributes significantly to the advancement of secure, intelligent IoT systems.

Keywords: AI, encoding, IoT, privacy, optimization, security

Introduction

The Internet of Things (IoT) ecosystem comprises interconnected physical entities equipped with software, sensors, and network connectivity, enabling seamless communication among devices and systems. IoT integration benefits industries such as manufacturing, healthcare, commerce, and transportation by enhancing decision-making, safety, customer experiences, and efficiency. However, it also introduces privacy and security challenges due to its complex and dynamic nature.

Privacy and security, though distinct, are closely linked. Security measures like cryptography protect data from breaches, but often require data collection, raising privacy concerns. Striking a balance is essential to ensure both data protection and individual autonomy. Rapid technological advancements amplify these challenges, necessitating adaptive strategies.

Traditional Intrusion Detection Systems (IDS) struggle to cope with modern threats, while AI-based IDS offer real-time anomaly detection. However, they require large datasets and are vulnerable to data manipulation. Protecting privacy in AI requires techniques like encryption, anonymization, and federated learning. Yet, encrypted training data can degrade accuracy due to added noise and computational cost.

This study introduces a unified AI-powered framework to address IoT privacy, security, and optimization. A novel technique, Replacement Encoding (RE), is proposed to anonymize data by substituting characters, symbols, and numbers with unique values, preserving feature relationships and enabling efficient data preprocessing. It supports the transformation of categorical to numerical data and the automatic conversion of unstructured data into structured, noise-free formats, while allowing original data retrieval.

The framework uses RE in AI-based IDS for real-time threat detection, ensuring data confidentiality and integrity. An optimization technique reduces computational load on resource-constrained IoT devices, maintaining high performance.

Key contributions include:

- An AI-based integrated mechanism addressing IoT privacy and security.
- The RE method for protecting sensitive data during AI training.
- Optimization techniques to enhance performance on low-resource devices.
- Conversion of categorical data into numerical form for seamless AI processing.
- Automated structuring and noise removal from unstructured data.
- Applications of RE in data protection, preprocessing, and AI models like GPT.
- Comprehensive analysis comparing the proposed approach with existing methods.

This framework marks a significant advancement in securing and optimizing the IoT ecosystem, supporting its future growth.

Literature Review

The literature review in the paper highlights the growing concerns of privacy and security in IoT systems, driven by the massive data collection and vulnerability of IoT devices. The review emphasizes:

Key Issues Identified:

1. Challenges in a Unified Mechanism:
    - Privacy and security have typically been treated separately.
    - Integrating them into a single mechanism is difficult due to the IoT's heterogeneity and dynamic nature.
2. Privacy vs. Security:
    - Privacy protects data from unauthorized access (via encryption, concealment, etc.).
    - Security defends against cyberattacks and data tampering (via IDS, detection systems, etc.).
    - While they overlap, their implementation goals can conflict.

3. Existing Encoding Techniques:
   - Common AI encoding methods like one-hot, ordinal, hash, and label encoding improve model performance but do not prioritize privacy.
   - These methods may increase dimensionality, cause data redundancy, and lack privacy-preserving capabilities.
4. AI's Role & Concerns:
   - AI helps in pattern recognition and intrusion detection, but training models on sensitive data raises privacy risks.
   - Unauthorized entities (data scientists, attackers) could misuse training data.

Proposed Solution:

- The authors propose a novel "Replacement Encoding" (RE) method designed to:
  - Anonymize sensitive data by replacing each character, symbol, and number with a unique value.
  - Maintain model accuracy and structure.
  - Enable automatic data preprocessing (including transforming categorical data to numerical form).
  - Allow lossless transformation, i.e., the original data can be recovered from encoded data.

Advantages Over Traditional Methods:

- Handles unstructured or noisy data by standardizing it.
- Simplifies feature analysis and improves AI training efficiency.
- Reduces computational overhead, suitable for resource-constrained IoT devices.
- The method also automates the data transformation process and works well for high-risk domains like finance and government.

Security Integration:

- RE is paired with AI-based intrusion detection systems (IDS) focused on anomaly detection.
- Outperforms traditional signature- and specification-based IDS in identifying new and evolving threats.

Supporting Tables:

- Table 1 compares RE with other encoding techniques.

Technique	Description	Preprocessing Efficiency	Performance Enhancement	Privacy Focus	Data Dimensionality	Invertible	Pros	Cons
One-Hot [22],[21] [23]	Converts categories to binary vectors	Low	No	No	High	Yes	Easy implementation, good for small categories	High dimensionality, computationally expensive for large categories
Label [21],[24], [23]	Assigns unique integers to categories	Moderate	Moderate	No	Moderate	Yes	Easy implementation, preserves order	Introduces unintended relationships
Binary [21], [25] ,[23],[26]	Converts categories to binary vectors	Moderate	Moderate	No	Moderate	Yes	More memory-efficient than one-hot, good for large categories	Less interpretable
Ordinal [27], [18] , [28][26]	Encodes based on inherent order	Low	Moderate	No	Moderate	Yes	Preserves order information	Sensitive to order changes
Count [29], [26] [30]	Converts categories to frequency counts	Moderate	Moderate	No	Low	Yes	Captures category frequency, efficient for large datasets	May not capture relationships
Target [31],[32], [33]	Uses target variable to assign numerical values	Moderate	High	No	Moderate	Not always	Captures relationship with target, efficient	Overfitting risk, sensitive to outliers
Hashing [34],[35], [36]	Converts categories to unique hash values	High	High	No	Low	Not always	Very efficient	Loses order information
Replacement (Proposed)	**Replaces characters/symbols and numbers with unique values**	**High**	**High**	**Yes**	**Low**	**Usually,**	**Improved privacy, efficient, constant format, easy reversion**	**Small memory buffer needed**

- Table 2 classifies types of IDS (signature, anomaly, specification).

Type	Description	Advantages	Disadvantages
Signature-based	Compares network traffic or device behavior to known attack signatures.	Fast and efficient for known attacks. - Easy to implement.	Vulnerable to zero-day attacks. - Requires frequent signature updates.
Anomaly-based	Identifies deviations from established normal behavior patterns.	- Can detect novel attacks. - Adapts to changing environments.	High false positive rates. Requires accurate baseline establishment.
Specification-based	Analyzes traffic against predefined security protocols and specifications.	Enforces compliance with security standards. - Catches protocol violations.	Limited to specific protocols. - May not detect novel attacks.

- Table 3 lists traditional privacy-preserving methods (like Homomorphic Encryption, Federated Learning), noting they are good for network communication but not dataset-level AI privacy.

Encoding Technique	Description	Advantages	Disadvantages
Homomorphic Encryption	Allows computations on encrypted data without decryption	Preserves privacy during data processing	High computational overhead
Attribute-Based Encryption	Encrypts data such that only users with specific attributes can decrypt it	Fine-grained access control	Complex key management
Differential Privacy	Adds random noise to data to protect individual privacy	Provides a strong privacy guarantee	Reduces data accuracy
Lightweight Encryption	Utilizes simpler algorithms for resource-constrained IoT devices	Efficient for IoT devices with limited resources	Potentially weaker security compared to traditional methods
Elliptic Curve Cryptography (ECC)	Provides high security with smaller key sizes compared to RSA	Efficient for resource-constrained devices	More complex implementation
Secure Multiparty Computation (SMC)	Allows parties to jointly compute a function over their inputs while keeping those inputs private	Enables collaborative computations without data sharing	High computational and communication complexity
Pseudonymization	Replaces private data with pseudonyms	Reduces risk of re-identification	Requires secure management of pseudonym mappings
Blockchain-Based Encryption	Uses blockchain to enhance data integrity and security	Tamper-resistant and decentralized	Scalability and performance issues
Data Masking	Hides original data with modified content	Maintains format and usability of data	May not provide strong security
Secure Sockets Layer (SSL)	Encrypts data in transit to prevent interception	Widely adopted, provides end-to-end security	Does not protect data at rest or during processing
[Replacement Encoding]	**RE is Lightweight preprocessing gateway used to camouflage data and protect privacy.**	**Camouflage original data, protect data during processing, Replaces original data with not arbitrary values. Efficient for constrained IoT devices. Easy key management**	**Vulnerable to AI cryptanalysis, Increased digits (>3) for encoding can cause collisions.**

Methodology

PROPOSED MECHANISM FOR PRIVACY AND SECURITY WITH REPLACEMENT ENCODING

In the IoT domain, sensitive data is frequently vulnerable to unauthorized access and observation. This information, which can include data from security, finance, military, and government entities, is highly confidential. To tackle this critical concern, the present study proposes a novel approach known as replacement encoding. This technique aims to safeguard sensitive information by transforming it into a synthetic form (encoded data) with a consistent and constant format and dimensionality. The core principle of replacement encoding involves substituting each original character (letters, symbols, numbers) with a unique, non-arbitrary value. This value assignment begins with a secret key and sequentially continues throughout the data. This approach utilizes three distinct secret keys to enhance security, rendering the encoded data meaningless to humans while retaining its utility for AI models and machines. Crucially, only the data owner possesses the ability to perform replacement encoding, as illustrated in Fig. 1. This ensures complete control over sensitive information, preventing unauthorized entities from viewing or accessing the original data. Data scientists, entrusted with the encoded dataset, can train AI models without exposing the underlying information. Once trained, the model is delivered back to the data owner for testing through a secure replacement encoding gateway. This closed-loop system guarantees that only the data owner has access to the original information, safeguarding its privacy and confidentiality. Fig. 1 visually depicts the workflow of replacement encoding, highlighting its role in data privacy and AI-powered security detection. The diagram clearly demonstrates how data is encoded for privacy, utilized for training the AI model, and ultimately tested through a secure gateway, all while maintaining complete control over the sensitive information by the data owner. This innovative approach offers a robust and efficient solution for safeguarding sensitive data within the IoT environment. By leveraging replacement encoding, organizations can effectively secure their data and unleash the full potential of AI-driven security solutions.

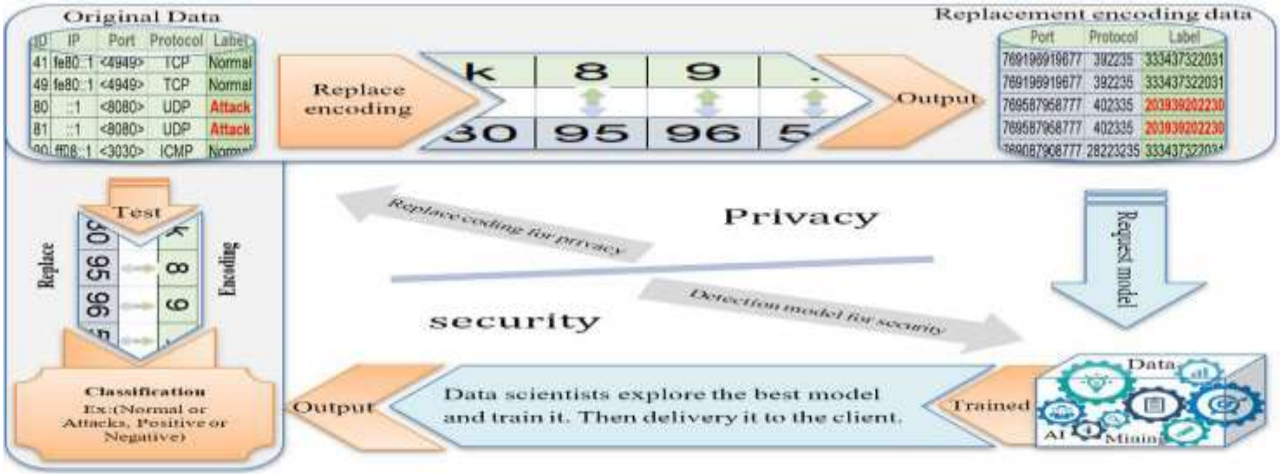


FIGURE 1. The proposed integrated privacy and security approach in an

AI-based IoT ecosystem (Replacement encoding for privacy and AI for detection).

Letters	A a	B b	C c	D d	E e	F f	G g	H h	I i	J j	K k	L l	M m	N n	O o	P p	Q q	R r	S s	T t	U u	V v	W w	X x	Y y	Z z

Symbols	?	.	,	;	:	'	"	(	)	{	}	[	]	*	/	%	$	&	@	#	!	\	~	_	`

Math symbols	+	-	×	÷	=	<	>	≠	⩽	⩾	π	√	∞	∛	$\vert$	^

Numbers	0	1	2	3	4	5	6	7	8	9

FIGURE 2. Letters, symbols, and numbers that change according to encoding value.

Fig. 2 visually represents the letters, symbols, and numbers utilized in the English language. While numbers and symbols typically maintain consistency across various languages, the letters themselves may exhibit variation. This study exclusively employs the replacement encoding approach within the context of the American Standard Code for Information Interchange (ASCII). However, the replacement encoding technique can be adapted to encode data according to specific language types and contextual requirements, following the same procedural steps applied in the case of the English language.

Secret key: 20

Letters	A a	B b	C c	D d	E e	F f	G g	H h	I i	J j	K k	L l	M m	N n	O o	P p	Q q	R r	S s	T t	U u	V v	W w	X x	Y y	Z z
Replace-Coding	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45

Secret key: 20

Symbols	?	.	,	;	:	'	"	(	)	{	}	[	]	*	/	%	$	&	@	#	!	\	~	_	`
Replace-Coding	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70

Secret key: 20

Math symbols	+	-	×	÷	=	<	>	≠	⩽	⩾	π	√	∞	?	$\vert$	^
Replace-Coding	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86

Secret key: 20

Numbers	0	1	2	3	4	5	6	7	8	9
Replace-Coding	87	88	89	90	91	92	93	94	95	96

FIGURE 3. Assigned values for letters, symbols, and numbers with assumed secret key.

Fig. 3. Each secret key serves a specific purpose: (1) Secret Key 1: Initializes the replacement values for letters, beginning with a unique value and sequentially assigning subsequent values to each letter. (2) Secret Key 2: Initializes the replacement values for symbols, employing a separate value range distinct from the letter key and continuing the sequence for each symbol. (3) Secret Key 3: Initializes the replacement values for numbers, utilizing another distinct value range and sequentially assigning values to each number. Critically, these three secret keys operate independently and never overlap. This design ensures that the initial values for each character type are unique and independent, thus preventing attackers from gaining insights into the overall encoding scheme. For instance, if the secret key for letters starts with 20, the second key for symbols should begin at least at 20 + the number of letters or higher, and similarly, the third key for numbers should start at 20 + the number of letters + the number of symbols or

higher. This multi-key approach significantly increases the difficulty for attackers to decipher the replacement encoding scheme. With multiple independent keys operating across different character types, guessing the correct values becomes vastly more complex, especially when the key ranges are intentionally separated. Therefore,

ID	IP	Port	Protocol	Label
41	fe80::1	<4949>	TCP	Normal
49	fe80::1	<4949>	TCP	Normal
80	::1	<8080>	UDP	Attack
81	::1	<8080>	UDP	Attack
90	ff08::1	<3030>	ICMP	Normal

ID	IP	Port	Protocol	Label
9188	25249587505088	769196919677	392235	333437322031
9196	25249587505088	769196919677	392235	333437322031
9587	505088	769587958777	402335	203939202230
9588	505088	769587958777	402335	203939202230
9687	25258795505088	769087908777	28223235	333437322031

FIGURE 4. Original characters with assigned replacement value.

Fig. 4 illustrates the original characters in correspondence with their respective assigned

values within the groups of letters, symbols, and numbers.

Original characters Match																								
	f	e	t	c	p	u	d	i	m	n	o	r	a	l	k	0	1	3	4	8	9	:	<	>
Replace characters Match																								
	25	24	39	22	35	40	23	28	32	33	34	0	20	31	30	87	88	90	91	95	96	50	76	77

FIGURE 5. Original dataset converted to a replacement encoding dataset.

Fig. 5 illustrates the transformative effect of replacement encoding. On the left side of Fig. 5, the original data is presented in its initial format, while the right side depicts the corresponding encoded dataset.

ID	IP	Port	Protocol	Label
9188	25249587505088	769196919677	392235	333437322031
9196	25249587505088	769196919677	392235	333437322031
9587	505088	769587958777	402335	203939202230
9588	505088	769587958777	402335	203939202230
9687	25258795505088	769087908777	28223235	333437322031

ID	IP	Port	Protocol	Label
41	fe80::1	<4949>	TCP	Normal
49	fe80::1	<4949>	TCP	Normal
80	::1	<8080>	UDP	Attack
81	::1	<8080>	UDP	Attack
90	ff08::1	<3030>	ICMP	Normal

FIGURE 6. Reverting encoding dataset to the original form.

Fig. 6 presents a mapping of original characters to their corresponding replacement values. Original characters assigned with replacement values and it facilitate the straightforward reversal of encoded data to its original form. This not only ensures data integrity but also enables seamless analysis or manipulation if required.

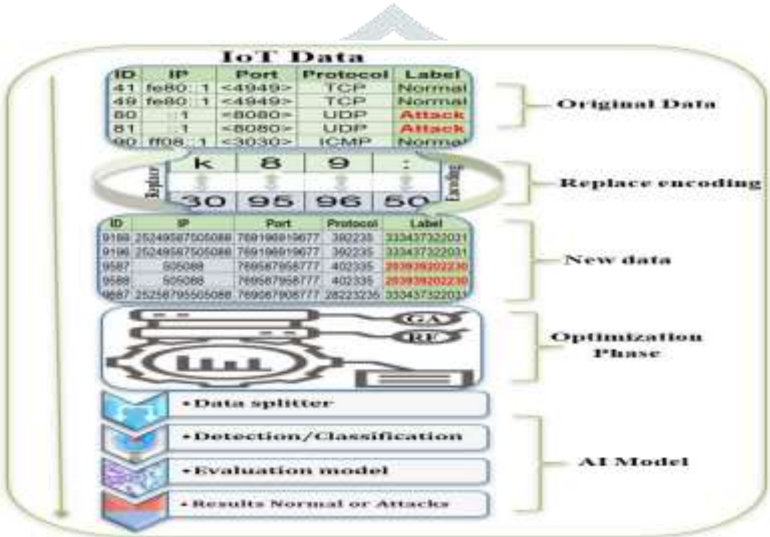
Pseudocode: IDS based AI with Replacement encoding

```
Secret key_for_letters= secret value_1;
Secret key_for_numbers = secret value_2;
Secret key_for_Symbols = secret value_3;
Assign values for each Letter start with secret value_1;
Assign values for each Number start with secret value_2;
Assign values for each Symbol start with secret value_3;
Inputs (Original dataset);
Replace (Letters, Numbers, Symbols) via new assigned values;
Generated a new dataset with Replacement coding values;
Optimization_techniques_for selection (GA);
Split_data (train (80%), test (20%));
Detection_Classifiers (ML_RF, DL_DNN);
Outputs (Class label, Probability of class);
Prevention (Class label="Attacks && Probability >90" go to 'stop' else 'continue');
```

FIGURE 7. Pseudocode for IDS-based AI with Replacement encoding.

Therefore, replacement encoding presents a potent and versatile tool for bolstering data privacy and enhancing AI performance. By leveraging its unique capacity to uphold data consistency, preserve dimensionality, convert data into numerical format, automate preprocessing, and facilitate data reversion, this approach holds immense potential for researchers and practitioners across various domains aiming to optimize their AI models and safeguard sensitive infor- mation. The pseudocode for an IDS-based AI utilizing the proposed replacement encoding approach is illustrated in Fig. 7.

Overall, integrating Replacement Encoding (RE) with AI models for real existing IoT systems enhances privacy, security, and performance. IoT practitioners and developers should explore the compatibility and adaptability of the RE mechanism with their specific systems to reinforce privacy and security measures while improving overall system perfor- mance. However, it is crucial to optimize RE implementation for resource-constrained devices to balance functionality with resource limitations. To implement the RE mechanism in real-world IoT systems, the procedures and guidelines shown in Fig. 7 can be considered.



In this study, replacement encoding has been proposed as a promising approach to enhancing both data privacy and AI model performance in scientific research. It offers several advantages over traditional data encoding techniques, making it particularly suitable for applications involving sensitive data, including security, personal information, and military intelligence, as mentioned earlier. The visual representation of the replacement encoding process, as applied to an anomaly detection system designed for identifying attacks in an IoT network, is depicted in Fig. 9. The proposed system for detecting anomalies within the IoT network can be summa- rized as follows: In the initial stage, the original data consists of a combination of categorical and numerical values, along with potentially unreadable symbols and tags. The dataset used in this case encompasses data generated by various.

IoT devices within the network. To ensure secure encoding, three distinct secret keys are generated, each corresponding to a specific character type within the data: letters, sym- bols, or numbers. During the replacement encoding phase, every character, symbol, and number in the original data is substituted with its corresponding value derived from the respective secret key. This encoding process results in a transformed dataset that maintains a consistent and constant format, enabling efficient processing by AI models.

The optimization phase is crucial as it ensures that the AI models are trained on the most suitable data. In this study, genetic algorithms are employed to enhance the performance and accuracy of the models used for anomaly detection within the IoT network.

After encoding and optimization, the encoded dataset undergoes processing by AI models specifically chosen for their effectiveness in anomaly detection tasks. Models that demonstrated high efficiency were selected for implementa- tion in this experiment. Random Forest is implemented from machine learning, while deep neural networks are employed from deep learning in order to classify attacks within the IoT network. To establish a prevention threshold, the model scores obtained from metrics such as class label

(CL) and the probability of class (PC) are utilized.

In this study, a threshold of 0.9 has been identified as the optimal value for the intrusion prevention system (IPS). According to this threshold, the system operates as follows: if the class label (CL) is ‘‘attack’’ and the probability of class (PC) is greater than or equal to 0.9, the system triggers an alert to ‘‘stop,’’ effectively halting the malicious activity. Similarly, if the class label (CL) is ‘‘normal’’ and the probability of class (PC) is greater than or equal to 0.9, the system issues an alert to ‘‘continue,’’ allowing normal behavior to persist. In all other instances, the system takes action to detect new attacks, update the dataset, and remove outdated records, ensuring adaptability and reducing the burden on IoT devices. This optimized approach enables the proposed mechanism to effectively distinguish between normal and malicious behav- ior within the IoT network. By permitting normal activity to proceed while effectively thwarting all attacks, the system significantly contributes to bolstering data privacy and secu- rity within the network.

I. DATA COLLECTION

For the purpose of this experiment and study, the dataset selection was conducted with utmost care. The chosen dataset, known as the CICIoT2023 dataset [47], offers a valuable resource for researchers and developers in the field of IoT security. Its real-time nature, diverse attack coverage, and publicly available format make it a valuable asset for advancing research and development in this critical area. It has been deemed suitable for utilization with AI models. In this study, our objective is to employ and evaluate proposed replacement encoding approaches on the CICIoT2023 dataset.

In this paradigm, the fitness function operates on two fronts: Performance, exemplified by accuracy, and Relative Importance, which delineates the significance of features. GA’s pursuit culminates when the fitness function attains predefined thresholds—Accuracy ≥90% and Importance ≥90%—signifying the attainment of the optimal solution.

To dynamically adapt to evolving data, the number of selected features undergoes adaptive modulation, thereby streamlining the iterative process. The cumulative relative importance of correlation coefficients (CC) for significant features must surpass 0.9, while for insignificant features, it must remain below 0.1. Ensuring coherence, the sum of relative importance among any two points adheres rigorously to unity. Furthermore, our tailored measures validate GA’s efficacy, yielding coefficients of 0.97 and 0.003, underscoring its prowess in selecting the most optimal features, as illustrated in Fig.

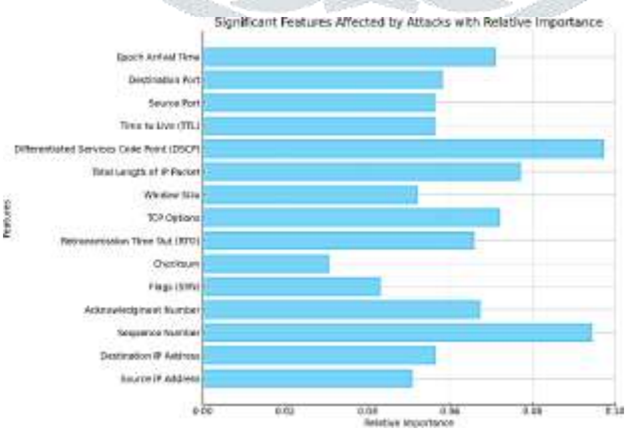


FIGURE 8. Top 15 significant feature selection using genetic algorithm.

The GA algorithm was employed to automatically select the most relevant features from the derived datasets. Its objective was to attain an optimal solution by identifying significant parameters and determining the highest correla- tion features [50]. To control the threshold for the optimal solution, we established a criterion whereby the accuracy must surpass 90% in order to halt the genetic algorithm. The genetic algorithm successfully identified and selected the top 15 features with the highest correlations for utilization in the classification process. Continuing from the previous scenario, the integration of the proposed replacement encoding with an RF classifier and GA optimization has yielded remarkable results.

T=9392221	DDoS	DoS	Recon	Web	BruteF	Spoofing	Mirai	Benign
DDoS	6794208 72.34%	13703 0.15%	13 0.00%	702 0.01%	311 0.00%	0 0.00%	45023 0.48%	25034 0.27%
DoS	705 0.01%	1375617 14.65%	708 0.01%	146 0.00%	59 0.00%	17522 0.19%	33490 0.36%	7043 0.07%
Recon	311 0.00%	50002 0.53%	68159 0.73%	189 0.00%	189 0.00%	0 0.00%	0 0.00%	0 0.00%
Web	1067 0.01%	7567 0.08%	17 0.00%	3547 0.04%	177 0.00%	0 0.00%	12987 0.14%	21054 0.22%
BruteF	3 0.00%	10345 0.11%	22 0.00%	0 0.00%	1535 0.02%	0 0.00%	0 0.00%	0 0.00%
Spoofing	0 0.00%	4 0.00%	105 0.00%	17 0.00%	0 0.00%	59410 0.63%	0 0.00%	0 0.00%
Mirai	617 0.01%	160897 1.71%	886 0.01%	351 0.00%	244 0.00%	9356 0.10%	423293 4.51%	42098 0.45%
Benign	1 0.00%	12 0.00%	1003 0.01%	13 0.00%	97 0.00%	11012 0.12%	12031 0.13%	179319 1.91%
Summary	6796912 99.96% 0.04%	1618147 85.01% 14.99%	70913 96.12% 3.88%	4965 71.44% 28.56%	2612 58.77% 41.23%	97300 61.06% 38.94%	526824 80.35% 19.65%	274548 65.31% 34.69%

FIGURE 9. RF results of confusion matrix of replacement encoding with selective 15 features using GA

This integrated approach has yielded an exceptional over- all accuracy of 94.81%, accompanied by a remarkably low misclassification rate of 0.05%. In contrast, the performance of the DNN classifier, with an accuracy of 90.16%, was surpassed. Fig. 26 visually presents the confusion matrix for the top-performing RF classifier, providing a comprehensive overview of the classification results. Furthermore, Fig. 27

presents detailed performance metrics, including accuracy, F1-score, precision, and recall, tailored to each attack type.

Attacks	Precision	1-Precision	Recall	1-Recall	f1-score
DDoS	0.9877	0.0123	0.9996	0.0004	0.9936
DoS	0.9584	0.0416	0.8501	0.1499	0.901
Recon	0.5735	0.4265	0.9612	0.0388	0.7184
Web	0.0764	0.9236	0.7144	0.2856	0.1381
BruteF	0.1289	0.8711	0.5877	0.4123	0.2115
Spoofing	0.9979	0.0021	0.6106	0.3894	0.7576
Mirai	0.6637	0.3363	0.8035	0.1965	0.727
Benign	0.8812	0.1188	0.6531	0.3469	0.7502
Accuracy			0.9481		
Misclassification Rate			0.0519		
Macro-F1			0.6497		
Weighted-F1			0.9504		

FIGURE 10. The details of performance parameters for highest classifier (RF) in each attacks with replacement encoding and selective features using GA.

It is important to highlight that this approach demonstrates significant improvement compared to results obtained in previous scenarios, emphasizing the effectiveness and supe- riority of the proposed combined strategy. Applying a genetic algorithm for feature selection, aimed at minimizing the num- ber of features and selecting the most relevant ones, yielded significantly improved results compared to the previous sce- nario relying on traditional methods.

FIGURE 11. The overall performance for RF and DNN classifiers with only numerical data and 52 features.

A. *OVERALL ACCURACY*

Figures 28, 29, and 30 respectively illustrate the overall performance metrics of both RF and DNN classifiers across different approaches. Fig. 28 summarizes the performance metrics when using only numerical data and 52 features. Fig. 29 displays the performance of the RF and DNN classifiers with all 100 features using replacement encod- ing. Finally, Fig. 30 presents the performance metrics with replacement encoding and features selections optimized to 15 features using GA. The overall accuracy of the best classi- fiers across all scenarios is depicted in Fig. 31. Notably, when solely applying numerical data (52 attributes) to the derived

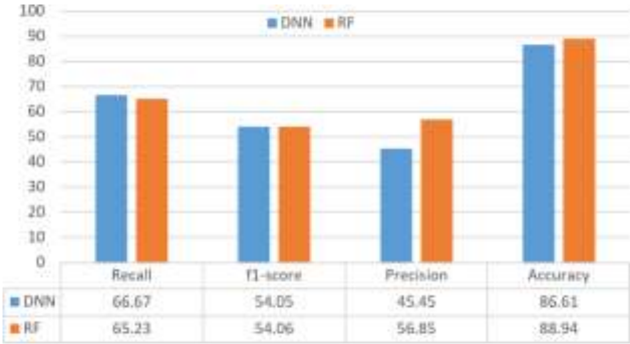


FIGURE 12. The overall performance for RF and DNN classifiers with full featuers via replacment encoding.

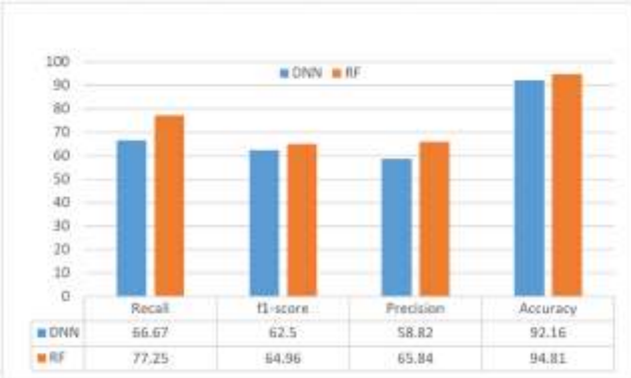


FIGURE 13. The overall performance for RF and DNN classifiers with replacement encoding and selective features using GA.

dataset, the DNN classifier achieved an impressive overall accuracy of 91.54%. Conversely, when integrating our pro- posed replacement encoding with RF classifiers and utilizing the full set of features (100 features derived from PCAP files), the overall accuracy reached 88.94%. This indicates that the scenario involving only numerical data outperformed the one with the proposed replacement (RE) encoding integrated with AI classifiers. The reason behind this discrepancy lies in the proposed replacement encoding scenario, where we imple- mented the full set of features. However, given the variability in the effects of attacks, it is crucial to consider all features to accurately gauge the performance of AI models.

In the initial scenario, we were unable to select the full set of features due to the presence of mixed categorical and numerical data in the derived datasets. To mitigate potential failures in AI model processing, we specifically utilized only numerical data. However, with the implemen- tation of the proposed replacement encoding, all features were selected, as they were transformed into numerical data, facilitating easier processing by the AI models. Furthermore, upon integrating our proposed replacement encoding with AI classifiers and combining it with GA algorithms for opti- mizations, the RF classifier achieved an outstanding overall accuracy of 94.81%. This underscores the effectiveness and substantial improvement attained through the integration of

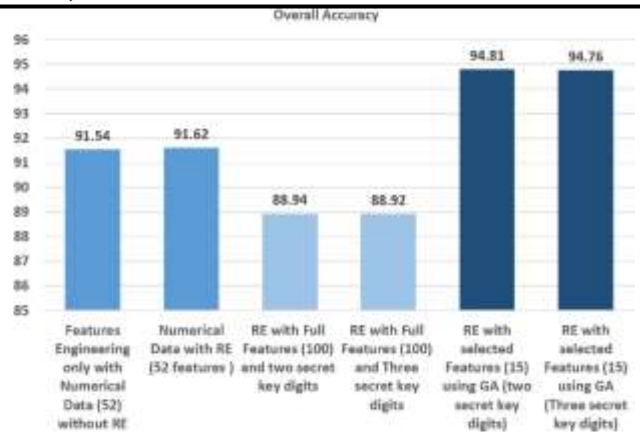


FIGURE 14. The summary of highest results for only numerical data features, replacement encoding with full features, and replacement encoding with selected features using GA. our proposed encoding technique with AI classifiers and GA algorithms.AI models generally perform better with numerical data compared to categorical data. Choosing the right features significantly impacts classifier performance. Additionally, simply reducing features might not always lead to better results because anomalies can manifest differently across attacks, affecting different features in the data. For instance, a log anomaly in one attack might primarily affect the ‘‘time’’ feature, while another attack might target the ‘‘flags value’’ feature, and so on. Therefore, intelligent feature selection techniques, such as those mentioned earlier in this study (e.g., GA), offer a valuable approach. These techniques go beyond traditional methods by considering the relationships and connections between features (feature correlation and association). This allows them to identify the most influential features that capture anomalies effectively.

In the proposed RE approach, it is recommended that a secret key range consisting of two digits be utilized to ensure accurate restoration of the data to its original form. It is important to avoid duplicating the mapping of more than one number, letter, or symbol to a single value. In addition to avoiding overlapping of the secret key for numbers, letters, and symbols. It is advisable to choose a secret key comprising two digits, starting from 10 and not exceeding three digits (<100). However, in the implementation of the replacement encoding approach, three digits were used, and it yielded the same results, albeit with variations in processing time. While the experiment demonstrated successful implementa- tion with a three-digit key, achieving similar accuracy results to that of a two-digit key, it has led to increased processing time compared to the latter. While the replacement encoding method reveals the actual performance by processing cat- egorical data, including even high-impact anomalies, these features can potentially negatively impact accuracy. However, this approach offers a more realistic depiction of model per- formance. Moreover, it may yield positive effects through automated preprocessing and noise removal.

CONCLUSION

This study focused on exploring encoding mechanisms from the perspective of data privacy amidst the widespread prolif- eration of AI models across diverse domains such as security, IoT, and chat systems. Its objective was to safeguard sensi- tive data from unauthorized access by entities leveraging AI models. The study underscored the significance of encod- ing mechanisms not only in data pre-processing but also in upholding data privacy within AI models.

A novel privacy mechanism, termed Replacement Encod- ing (RE), was introduced. It is specifically designed to anonymize and camouflage sensitive data during AI model training, while also preserving the integrity and utility of the trained models. Furthermore, RE facilitates the automatic processing of data. The proposed approach integrates AI classifiers for anomaly detection in IoT networks and utilizes GA for performance improvement, offering a comprehensive solution to the challenges faced by IoT devices. The effective- ness of the RE approach in transforming data and features into a suitable format for AI models was demonstrated.

The feasibility of employing the RE approach was show- cased in various applications involving sensitive information such as security, banking, and personal data, alongside the utilization of GPT. The study investigated the impact of feature selection and data encoding on AI classifier perfor- mance for anomaly detection in IoT networks. While utilizing only numerical data (52 features) achieved high accuracy (91.54%) using the DNN classifier, the proposed RE with RF classifiers on the full feature set (100 features) (88.94%) high- lighted the importance of considering all features to capture diverse attack effects and provide actual performance for AI models. Furthermore, the integration of the RE mechanism with GA for optimization significantly improved accuracy (94.81%) using the RF classifier.

This research unveils the significant potential of the proposed mechanism in safeguarding data privacy and offers a fresh perspective on encoding techniques to ensure anonymity for machine-processed information. This is par- ticularly crucial in the current landscape of continuous advancements and widespread adoption of AI tools.

One potential limitation of the RE mechanism is the possi- bility that future AI algorithms may be able to detect patterns and discover the assigned values for each letter, number, and symbol in the data. This arises due to the frequent and sequential assignment of values throughout the entire dataset. Therefore, it is envisioned that refinements of the RE mechanism will be necessary to actively prevent the identification of patterns within the data by AI systems. These refinements aim to address potential vulnerabilities that could be exploited by malicious actors in the future, ensuring the sustained effectiveness and robustness of the mechanism in safeguarding privacy and security.

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